Mean Shift Clustering

A Non-Parametric Clustering Algorithm

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Outline

Historical Context

The paper about Mean Shift was first introduced by **Fukunaga and Hostetler in 1975** with the title *"The Estimation of the Gradient of a Density Function, with Applications in Pattern Recognition"* where they presented the mean shift algorithm as a method for finding modes of a density function given discrete data sampled from that function.

Later on in 2002, **Comaniciu and Meer** popularized the algorithm in their paper *"Mean Shift: A Robust Approach Toward Feature Space Analysis"* by demonstrating its effectiveness in computer vision tasks such as image segmentation and tracking.

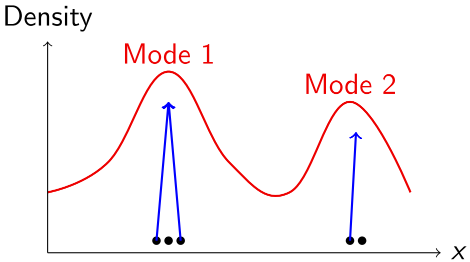
What is Mean Shift Clustering?

Core Idea:

- Find modes (peaks) of data density
- Points converge to nearest mode
- No need to specify number of clusters
- Density-based approach

Key Characteristic:

- Non-parametric
- Discovers clusters naturally
- Based on kernel density estimation

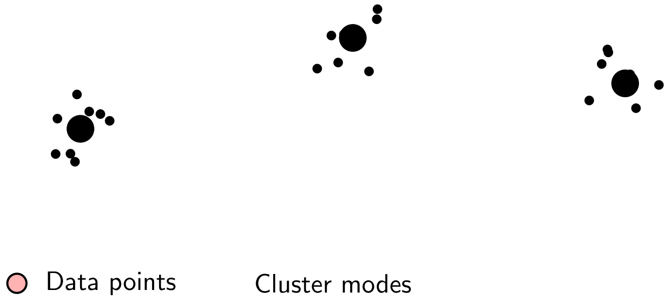


Points shift toward density peaks

Visual Intuition

Think of it as:

1. Imagine data points as balls on a hilly landscape
2. Each ball rolls uphill to the nearest peak
3. Balls ending at the same peak = same cluster



Advantage: Discovers cluster count automatically from data!

The Mathematics (Brief)

1. Kernel Density Estimation: Given n data points $\{x_1, x_2, \ldots, x_n\}$, the kernel density estimate at point x is:

$$\hat{f}(x) = \frac{1}{nh} \sum_{i=1}^{n} K\left(\frac{x - x_i}{h}\right)$$

where we use a **symmetric kernel**: $K(x) = c_k \cdot k(x^2)$

- $k : \mathbb{R}^+ \to \mathbb{R}$ is the **kernel profile** (function of squared distance)
- $h > 0$ is the bandwidth parameter
- c_k is a normalization constant
- x_i are the data points and no need to be inside the bandwidth

Key insight: If we set h too large, we may suffer from over-smoothing.

From KDE to Mean Shift

2. Deriving the Mean Shift Vector:

To find density modes, we take the **gradient** of $\hat{f}(x)$ and set it to zero.

Define the **shadow kernel profile**: $g(s) = -k'(s)$ (negative derivative of k)

Taking $\nabla \hat{f}(x) = 0$ and rearranging yields:

$$x = \frac{\sum_{i=1}^{n} x_i \, g\left(\left\|\frac{x-x_i}{h}\right\|^2\right)}{\sum_{i=1}^{n} g\left(\left\|\frac{x-x_i}{h}\right\|^2\right)}$$

Key insight: The function $g = -k'$ ensures we move **against** the gradient direction (toward increasing density). This weighted mean tells us where the mode is!

The Mathematics (Contd.)

3. Mean Shift Vector:

The difference between the weighted mean and current position:

$$m(x) = \underbrace{\frac{\sum_{i=1}^{n} x_i \, g\left(\left\|\frac{x-x_i}{h}\right\|^2\right)}{\sum_{i=1}^{n} g\left(\left\|\frac{x-x_i}{h}\right\|^2\right)}}_{\text{weighted mean using } g} - x$$

4. Update Rule:

$$x^{(t+1)} = x^{(t)} + m(x^{(t)})$$

Key insight: Since $g = -k'$, the mean shift vector always points toward increasing density (gradient ascent on the density surface).

Common Kernel Profiles

Let $s = \left\| \frac{x - x_i}{h} \right\|^2$ be the squared normalized distance.

Gaussian:

$$k(s) = \exp\left(-\frac{s}{2}\right)$$
$$g(s) = -k'(s) = \frac{1}{2}\exp\left(-\frac{s}{2}\right)$$

Epanechnikov:

$$k(s) = \begin{cases} 1 - s & s \leq 1 \\ 0 & \text{else} \end{cases}$$
$$g(s) = \begin{cases} 1 & s \leq 1 \\ 0 & \text{else} \end{cases}$$

Flat (Uniform):

$$k(s) = \begin{cases} 1 & s \leq 1 \\ 0 & \text{else} \end{cases}$$
$$g(s) = 0 \text{ (undefined at boundary)}$$

Why Epanechnikov?

- $g(s) = 1$ (constant) $\Rightarrow$ uniform weights
- Simple weighted mean of neighbors
- Most efficient computation
- Used by sklearn's MeanShift

Why not Flat?

Algorithm Steps

Algorithm 1 Mean Shift Clustering

Input: Data points $\{x_1, \ldots, x_n\}$, bandwidth h, kernel profile k, ϵ for convergence
Output: Cluster assignments

for each data point $\mathbf{x}_i$ **do**
    $\mathbf{y} \leftarrow \mathbf{x}_i$ ▷ Initialize at data point
    while not converged **do**
        Compute mean shift: $\mathbf{m}(\mathbf{y})$
        Update: $\mathbf{y} \leftarrow \mathbf{y} + \mathbf{m}(\mathbf{y})$
    end while
    Store converged point (mode) for $\mathbf{x}_i$
end for

Group points by their converged modes
Return cluster assignments

Advantages

1. **No predefined cluster count**
   - Automatically discovers number of clusters
   - Unlike K-means which requires k
2. **Flexible cluster shapes**
   - Can find non-spherical clusters
   - Follows natural density contours
3. **Single parameter: bandwidth h**
   - Controls granularity of clustering
   - Easier to tune than multiple parameters
4. **Robust to initialization**
   - Deterministic convergence to modes
   - No random initialization sensitivity

Limitations (and Why They Matter)

1. **Computational complexity:** $O(n^2 \cdot T)$ (T is the avg number of iterations to converge)
   - Why? Each point computes distance to all others at each iteration
   - Impact: Slow for large datasets ($n > 10{,}000$)
   - Mitigation: Use spatial indexing (KD-trees), binning, or sampling
2. **Bandwidth selection is critical**
   - Why? Too small $h \rightarrow$ many tiny clusters; too large $h \rightarrow$ all points in one cluster
   - Impact: Results highly sensitive to h
   - Mitigation: Cross-validation, adaptive bandwidth, or domain knowledge
3. **Curse of dimensionality**
   - Why? Density estimation becomes unreliable in high dimensions
   - Impact: Poor performance for $d > 10$
   - Mitigation: Dimensionality reduction (PCA, t-SNE) first

When to Use Mean Shift?

Good Scenarios:

- Unknown number of clusters
- Non-spherical cluster shapes
- Small to medium datasets
- Image segmentation
- Object tracking in video
- Mode-seeking tasks

Avoid When:

- Very large datasets ($n > 50{,}000$)
- High-dimensional data ($d > 10$)
- Real-time processing needed
- Clusters have similar densities
- Well-separated spherical clusters (use K-means instead)

Summary

Mean Shift Clustering:

- Core idea: Find density modes, points converge to nearest mode
- Strengths: No cluster count needed, flexible shapes, robust
- Weaknesses: $O(n^2)$ complexity, bandwidth sensitivity, high-dim issues

Experimental Findings:

- Outperforms K-means in cluster quality (synthetic & Iris)
- Requires dimensionality reduction for high-dim data (Wine)
- Excellent for image segmentation, but slower
- Bandwidth selection is critical

Best Use Cases:

- Small-medium datasets with unknown cluster count
- Non-spherical clusters or mode-finding tasks
- Image/video processing applications

References

1. Fukunaga, K., & Hostetler, L. (1975). *The estimation of the gradient of a density function, with applications in pattern recognition*. IEEE Transactions on Information Theory.
2. Comaniciu, D., & Meer, P. (2002). *Mean shift: A robust approach toward feature space analysis*. IEEE Transactions on Pattern Analysis and Machine Intelligence.

Thank You!

Questions?

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